Spatial Filtering & Image Enhancement

Histogram equalization, smoothing filters, sharpening filters, and gradient operators.

1. Intensity Transformation & Histogram Processing

Logarithmic and power-law (Gamma) transformations for contrast adjustment.

Histogram equalization: transformation function s = T(r) producing uniform PDF.

Local histogram statistics for adaptive image enhancement.

2. Spatial Filters

Smoothing spatial filters: box filter and Gaussian lowpass smoothing kernels.

Sharpening filters: Laplacian second-derivative operator for edge highlighting.

Unsharp masking and high-boost filtering for edge detail preservation.